General Bite #3: Propagation Basics + HF Skip and MUF

Sunspots, solar flux, K and A index, geomagnetic storms; then MUF, LUF, hop distances, and how skywave propagation actually works.

Groups G3A + G3B · 26 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Part 1: The Sun, The Earth's Magnetic Field, And HF (G3A)

Sunspots And Solar Flux

Sunspots are surface markers of solar activity. More sunspots = more ionizing UV/X-ray = denser ionosphere = better HF propagation, especially at higher frequencies.

Question	Answer	Q
Higher sunspot number means…	**Greater probability of good propagation at higher frequencies** (G3A01).	G3A01
Solar flux index (SFI) is…	**A measure of solar radiation at 10.7 cm wavelength** (G3A05). A daily proxy for solar activity.	G3A05
Least reliable bands at low solar activity	**15, 12, and 10 meters** (G3A04). High bands need ionization to refract; without sunspots the F2 critical frequency drops below them.	G3A04
20 m band supports worldwide daytime propagation…	**At any point in the solar cycle** (G3A07). 20 m is the workhorse all-cycle DX band.	G3A07

Geomagnetic Activity (K-index, A-index)

The Earth's magnetic field is constantly being buffeted by the solar wind. Disturbances are measured by the K-index (short-term) and A-index (long-term).

Question	Answer	Q
K-index measures…	**Short-term stability of Earth's geomagnetic field** (G3A12).	G3A12
A-index measures…	**Long-term stability of Earth's geomagnetic field** (G3A13). Daily summary value.	G3A13
What is a geomagnetic storm?	**A temporary disturbance in Earth's geomagnetic field** (G3A06).	G3A06
Effect of geomagnetic storm on HF	**Degrade high-latitude HF propagation** (G3A08). Polar paths get sandblasted with absorption.	G3A08
Benefit of high geomagnetic activity	**Creates auroras that can reflect VHF signals** (G3A09). The auroral oval becomes a VHF mirror.	G3A09
Coronal-hole solar wind effect	**HF communication is disturbed** (G3A14).	G3A14

Solar Flares And SIDs

Question	Answer	Q
SID effect on daytime ionosphere	**Disrupts signals on lower frequencies more than higher** (G3A02). D-region absorption is severe at low HF during a SID.	G3A02
How fast does flare radiation reach Earth?	**8 minutes** (G3A03). Light-speed photons.	G3A03
How fast does a coronal mass ejection (CME) reach Earth?	**15 hours to several days** (G3A11). Particle plasma, slow.	G3A11
Cause of the 26-28 day HF cycle	**Rotation of the Sun's surface layers around its axis** (G3A10). Active regions sweep past Earth every 26-28 days.	G3A10

Part 2: MUF, LUF, And HF Skip (G3B)

The Two Key Frequencies

MUF (Maximum Usable Frequency) = the highest frequency that will refract back to Earth on a given path (G3B08).
LUF (Lowest Usable Frequency) = the lowest frequency that will get through D-region absorption and reach the destination with usable signal (G3B07).
A working contact lives **between** the MUF and LUF. Above the MUF: signal punches through the ionosphere into space. Below the LUF: signal is absorbed before reaching the destination.

Question	Answer	Q
What affects the MUF?	Path distance/location, time of day/season, solar activity – **all of the above** (G3B02).	G3B02
Least-attenuation frequency for long-distance skip	**Just below the MUF** (G3B03). Higher frequency = less D-region loss; just under MUF maximizes signal.	G3B03

Behavior of waves between LUF and MUF	**Refracted back to Earth** (G3B05). The working zone.	G3B05
Behavior of waves below the LUF	**Attenuated before reaching the destination** (G3B06). D-region eats them.	G3B06
What happens when LUF exceeds MUF?	**Ordinary skywave propagation not possible over that path** (G3B11). The window closed.	G3B11

Skip-Hop Distances

Question	Answer	Q
Max one-hop distance via F2 region	**2,500 miles** (G3B09). The longest single-hop HF reach.	G3B09
Max one-hop distance via E region	**1,200 miles** (G3B10). Shorter because the E layer is lower.	G3B10
Short-path and long-path both arriving	**A slightly delayed echo might be heard** (G3B01). Long-path arrives noticeably later than short-path (classic ham reference: 1/7 second, around 100-150 ms depending on path).	G3B01
Determining propagation in real time	**Use a network of automated receiving stations on the internet** (G3B04). PSK Reporter, WSPRnet, Reverse Beacon Network.	G3B04

Summer-Daytime Low-Band Behavior

Typical of lower HF frequencies (80/160) during summer: High levels of atmospheric noise or static (G3B12). Tropical thunderstorms generate the QRN; summer noise makes 80/160 nearly unusable in daytime.

Mental Cheat Sheet

- Higher sunspot number → better HF, especially high bands.
- Low solar activity → 15/12/10 m are unreliable.
- 20 m works DX at **any** point in the cycle.
- SFI = 10.7 cm solar radio flux.
- K-index = short-term geomag stability.
- A-index = long-term geomag stability.
- Geomagnetic storm → degrades high-latitude HF.
- Geomag storms create auroras → VHF reflections.
- Solar flare radiation: 8 min to Earth.
- CME particles: 15 hr to several days.
- 27-day HF cycle = Sun's rotation.
- MUF below = refracted back, MUF above = punches through.
- Best DX freq = just below MUF.
- F2 one hop ≈ 2500 mi; E one hop ≈ 1200 mi.
- LUF > MUF: window closed.
- Long path delay = audible echo with short path.
- Summer 80/160 daytime = static dominated.

Common Traps

- SFI is **10.7 cm** solar flux, not "X-ray to radio flux ratio."
- K-index is **short-term**; A-index is **long-term**.
- Solar flare radiation arrives in **8 minutes**; CMEs in **days**. Don't swap them.
- 27-day cycle is **Sun's rotation**, not atmosphere or Moon.
- Best DX is **just below** MUF, not below LUF or below critical frequency.
- F2 hop is **2,500 mi**, not 12,000 (12,000 mi is just the pool's largest distractor – half Earth's circumference; actual EME path is 239,000 mi one way).
- E-region hop is **1,200 mi**, not 2,500.
- Long-path produces an **echo**, not periodic fading or 3 dB boost.
- Summer low-band issue is **atmospheric noise (static)**, not absorption.
- Geomag storm **degrades** high-latitude paths, doesn't improve them.

Takeaway: Higher sunspots = higher MUF = better high-band HF. The K-index/A-index pair report Earth's magnetic-field disturbance. Flares hit in 8 minutes, CMEs in days. MUF and LUF bracket the usable window for HF skywave; aim just below MUF for least attenuation. One-hop F2 ≈ 2500 mi, E ≈ 1200 mi.

Bite 3 Question Bank (G3A + G3B)

G3A01. How does a higher sunspot number affect HF propagation?
A. Higher sunspot numbers generally indicate a greater probability of good propagation at higher frequencies.
B. Lower sunspot numbers generally indicate greater probability of sporadic E propagation.
C. A zero sunspot number indicates that radio propagation is not possible on any band.
D. A zero sunspot number indicates undisturbed conditions.

G3A02. What effect does a sudden ionospheric disturbance have on the daytime ionospheric propagation?
A. It enhances propagation on all HF frequencies.
B. It disrupts signals on lower frequencies more than those on higher frequencies.
C. It disrupts communications via satellite more than direct communications.
D. None, because only areas on the night side of the Earth are affected.

G3A03. Approximately how long does it take the increased ultraviolet and X-ray radiation from a solar flare to affect radio propagation on Earth?
A. 28 days.
B. 1 to 2 hours.
C. 8 minutes.
D. 20 to 40 hours.

G3A04. Which of the following are the least reliable bands for long-distance communications during periods of low solar activity?
A. 80 meters and 160 meters.
B. 60 meters and 40 meters.
C. 30 meters and 20 meters.
D. 15 meters, 12 meters, and 10 meters.

G3A05. What is the solar flux index?
A. A measure of the highest frequency that is useful for ionospheric propagation between two points on Earth.
B. A count of sunspots that is adjusted for solar emissions.
C. Another name for the American sunspot number.
D. A measure of solar radiation with a wavelength of 10.7 centimeters.

G3A06. What is a geomagnetic storm?
A. A sudden drop in the solar flux index.
B. A thunderstorm that affects radio propagation.
C. Ripples in the geomagnetic force.
D. A temporary disturbance in Earth's geomagnetic field.

G3A07. At what point in the solar cycle does the 20-meter band usually support worldwide propagation during daylight hours?
A. At the summer solstice.
B. Only at the maximum point.
C. Only at the minimum point.
D. At any point.

G3A08. How can a geomagnetic storm affect HF propagation?
A. Improve high-latitude HF propagation.
B. Degrade ground wave propagation.
C. Improve ground wave propagation.
D. Degrade high-latitude HF propagation.

G3A09. How can high geomagnetic activity benefit radio communications?
A. Creates auroras that can reflect VHF signals.
B. Increases signal strength for HF signals passing through the polar regions.
C. Improve HF long path propagation.
D. Reduce long delayed echoes.

G3A10. What causes HF propagation conditions to vary periodically in a 26- to 28-day cycle?
A. Long term oscillations in the upper atmosphere.
B. Cyclic variation in Earth's radiation belts.
C. Rotation of the Sun's surface layers around its axis.
D. The position of the Moon in its orbit.

G3A11. How long does it take a coronal mass ejection to affect radio propagation on Earth?
A. 28 days.
B. 14 days.
C. 4 to 8 minutes.
D. 15 hours to several days.

G3A12. What does the K-index measure?
A. The relative position of sunspots on the surface of the Sun.
B. The short-term stability of Earth's geomagnetic field.
C. The short-term stability of the Sun's magnetic field.
D. The solar radio flux at Boulder, Colorado.

G3A13. What does the A-index measure?
A. The relative position of sunspots on the surface of the Sun.
B. The amount of polarization of the Sun's electric field.
C. The long-term stability of Earth's geomagnetic field.
D. The solar radio flux at Boulder, Colorado.

G3A14. How is long distance radio communication usually affected by the charged particles that reach Earth from solar coronal holes?
A. HF communication is improved.
B. HF communication is disturbed.
C. VHF/UHF ducting is improved.
D. VHF/UHF ducting is disturbed.

G3B01. What is a characteristic of skywave signals arriving at your location by both short-path and long-path propagation?
A. Periodic fading approximately every 10 seconds.
B. Signal strength increased by 3 dB.
C. The signal might be cancelled causing severe attenuation.
D. A slightly delayed echo might be heard.

G3B02. What factors affect the MUF?
A. Path distance and location.
B. Time of day and season.
C. Solar radiation and ionospheric disturbances.
D. All these choices are correct.

G3B03. Which frequency will have the least attenuation for long-distance skip propagation?
A. Just below the MUF.
B. Just above the LUF.
C. Just below the critical frequency.
D. Just above the critical frequency.

G3B04. Which of the following is a way to determine current propagation on a desired band from your station?
A. Use a network of automated receiving stations on the internet to see where your transmissions are being received.
B. Check the A-index.
C. Send a series of dots and listen for echoes.
D. All these choices are correct.

G3B05. How does the ionosphere affect radio waves with frequencies below the MUF and above the LUF?
A. They are refracted back to Earth.
B. They pass through the ionosphere.
C. They are amplified by interaction with the ionosphere.
D. They are refracted and trapped in the ionosphere to circle Earth.

G3B06. What usually happens to radio waves with frequencies below the LUF?
A. They are refracted back to Earth.
B. They pass through the ionosphere.
C. They are attenuated before reaching the destination.
D. They are refracted and trapped in the ionosphere to circle Earth.

G3B07. What does LUF stand for?
A. The Lowest Usable Frequency for communications between two specific points.
B. The Lowest Usable Frequency for communications to any point outside a 100-mile radius.
C. The Lowest Usable Frequency during a 24-hour period.
D. The Lowest Usable Frequency during the past 60 minutes.

G3B08. What does MUF stand for?
A. The Minimum Usable Frequency for communications between two points.
B. The Maximum Usable Frequency for communications between two points.
C. The Minimum Usable Frequency during a 24-hour period.
D. The Maximum Usable Frequency during a 24-hour period.

G3B09. What is the approximate maximum distance along the Earth's surface normally covered in one hop using the F2 region?
A. 180 miles.
B. 1,200 miles.
C. 2,500 miles.
D. 12,000 miles.

G3B10. What is the approximate maximum distance along the Earth's surface normally covered in one hop using the E region?
A. 180 miles.
B. 1,200 miles.
C. 2,500 miles.
D. 12,000 miles.

G3B11. What happens to HF propagation when the LUF exceeds the MUF?
A. Propagation via ordinary skywave communications is not possible over that path.
B. HF communications over the path are enhanced.
C. Double-hop propagation along the path is more common.
D. Propagation over the path on all HF frequencies is enhanced.

G3B12. Which of the following is typical of the lower HF frequencies during the summer?
A. Poor propagation at any time of day.
B. World-wide propagation during daylight hours.
C. Heavy distortion on signals due to photon absorption.
D. High levels of atmospheric noise or static.

Answer Key

G3A01: A	G3A02: B	G3A03: C	G3A04: D	G3A05: D
G3A06: D	G3A07: D	G3A08: D	G3A09: A	G3A10: C
G3A11: D	G3A12: B	G3A13: C	G3A14: B	G3B01: D
G3B02: D	G3B03: A	G3B04: A	G3B05: A	G3B06: C
G3B07: A	G3B08: B	G3B09: C	G3B10: B	G3B11: A
G3B12: D				